NATIONAL FLAGSHIP CONFERENCE · SECOND EDITION
SUSTAINX 2026
Circularity in Energy, Earth and Environment
CONFERENCE July 24 – 25, 2026 | WORKSHOP July 23, 2026 | VENUE IIT Madras, Chennai | WEBSITE sustainxconference.org
ORGANISED BY
PARTNERS & SPONSORS
POSTER ID: SUS-PP-21

# Energy System Transition as Carbon Valorisation: Grid Expansion, Fossil Displacement, and Decentralised Renewable Persistence in Rural India

Srikant Shaw[1,*], Santosh Kumar Sahu[1,2]

[1]Department of Humanities and Social Sciences —Indian Institute of Technology Madras, Chennai, India · [2]School of Sustainability, IIT Madras, Chennai, India

EMAIL hs23d008@smail.iitm.ac.in ID ORCID 0009-0006-5155-8719 * Corresponding author

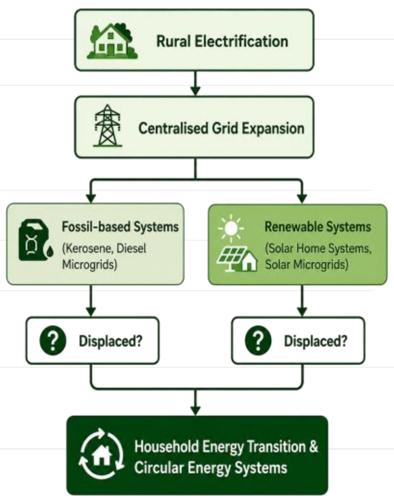
SCAN FOR MORE

## 01 INTRODUCTION

✓ India's rural electrification program has substantially expanded centralized grid access. While electrification improves household welfare, little is known about its interaction with decentralized energy technologies. We investigate whether grid expansion displaces all decentralized systems or selectively replaces fossil-based technologies while allowing renewable alternatives to persist. This distinction has important implications for circular energy transitions and long-term energy planning.

## 02 OBJECTIVES

- ✓ Estimate the effect of grid expansion on household energy transitions.
- ✓ Examine whether fossil and renewable decentralized systems respond differently.
- ✓ Assess implications for circular energy transitions.




## 03 METHODOLOGY

| Data | ACCESS Panel (2015–2018) |
|---|---|
| Sample Size | 17,635 households, 54 Districts, 6 Indian States |
| Identification | District-level Difference-in-Differences |
| Treatment | Above-median district grid expansion |
| Controls | Household Covariates |
| Robustness | Fixed Effects + First Difference (FD) + Augmented FD |

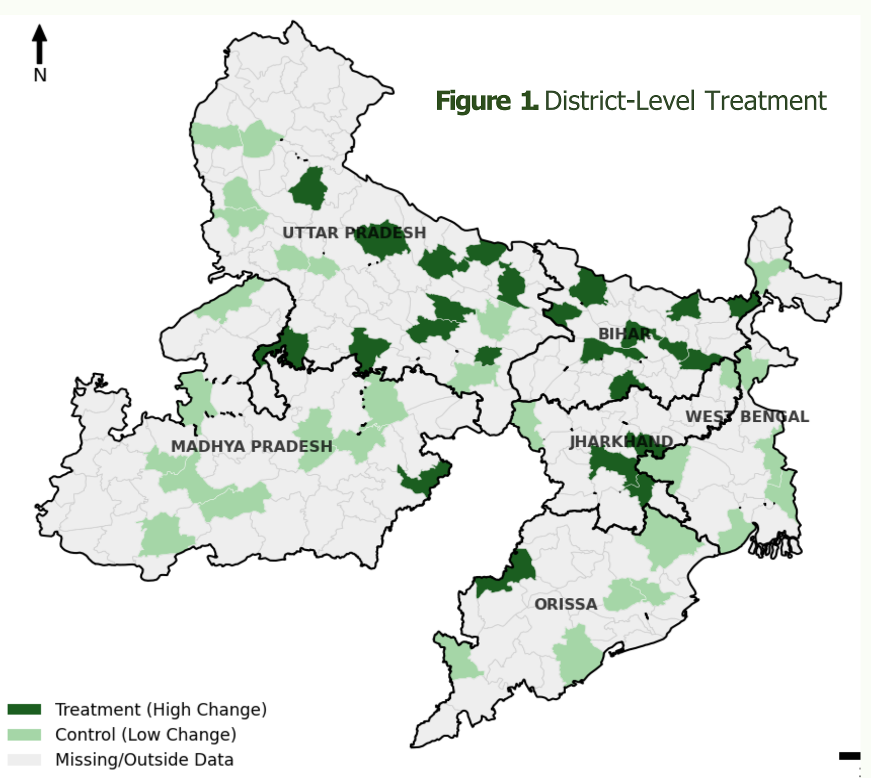



**Figure 1.** District-Level Treatment

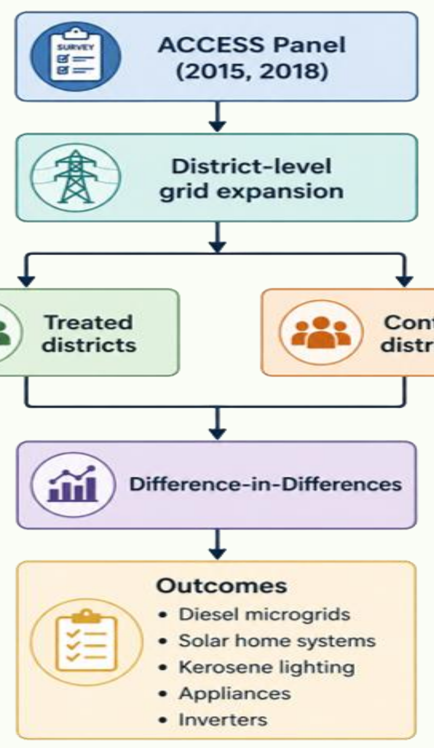



**Figure 2.** Schematic of the proposed methodology.

## 04 RESULTS & DISCUSSION

Fixed-effects Difference-in-Differences estimates. Positive values indicate improved access or welfare, while negative values indicate reduced dependence on fossil-based energy technologies. Error bars denote 95% confidence intervals.

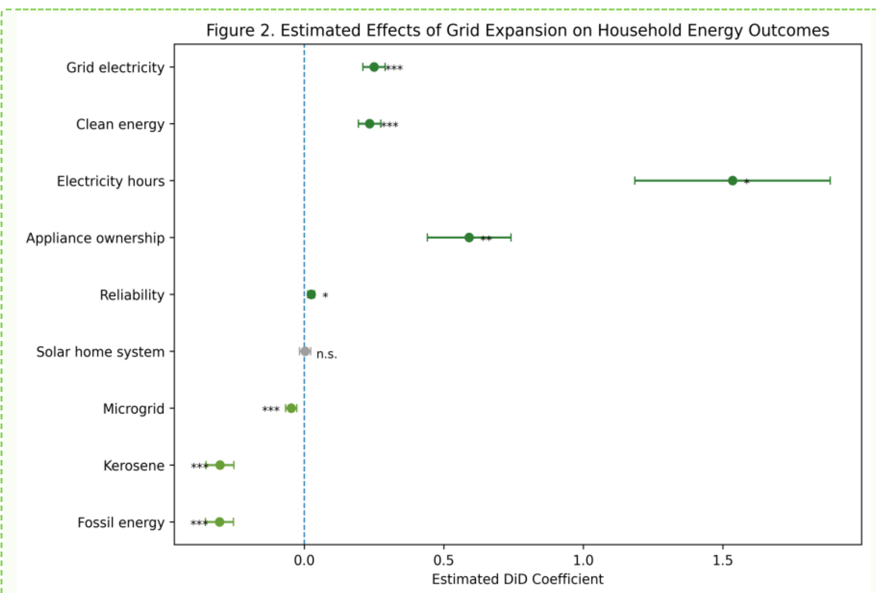
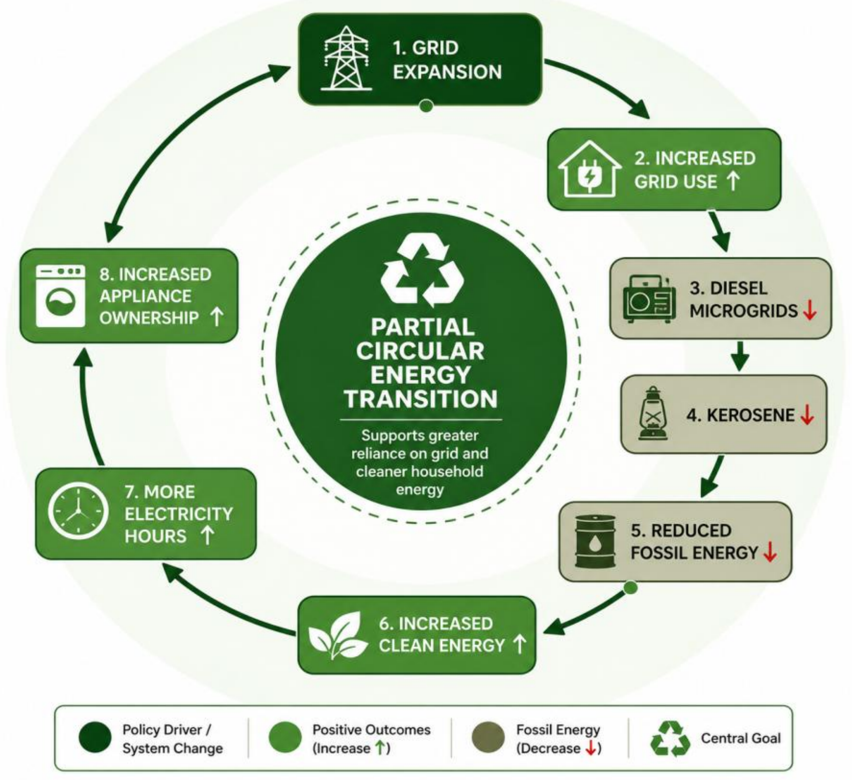



**Figure 3.** Estimated Effects of Grid Expansion on Household Energy Outcomes

**Table 1.** Main Treatment Effects.

| Outcome | Effect | Significance Level |
|---|---|---|
| Grid Electricity | + 25.0 pp | *** |
| Microgrid | - 4.7 pp | *** |
| Solar Lanterns/Home Systems | NS | --- |
| Inverter | + 2.5 pp | *** |
| Kerosene | - 30.2 pp | *** |
| Fossil Energy | - 30.4 pp | *** |
| Clean Energy | + 23.4 pp | *** |
| Electricity Hours | + 1.53 hrs | * |

*** p<0.01, ** p<0.05, * p<0.10, NS – Not Significant




**Scheme 1.** Mechanism of Household Energy Transition

District-level grid expansion functions as the catalyst for carbon valorisation by reshaping household energy portfolios. Instead of uniformly replacing decentralised systems, electrification selectively displaces fossil-based technologies while allowing renewable alternatives to persist. This transition reduces carbon-intensive energy use, improves electricity availability, and increases appliance adoption, thereby generating greater economic and social value from cleaner electricity use. The coexistence of centralised grids and decentralised renewables demonstrates that carbon valorisation can be achieved through complementarity, rather than complete technological substitution, thereby supporting a more circular and resilient rural energy system.

## 05 POLICY & IMPLICATIONS

**Key Finding 1: Selective Technology Displacement**

Grid expansion significantly reduced diesel microgrids and kerosene use but had no statistically significant effect on solar home systems.

**Key Finding 2: Welfare Improvements**

Households experienced longer electricity supply and increased ownership of electrical appliances, indicating improved energy services.

**Key Finding 3: Circular Energy Transition**

Electrification reduced fossil dependence but did not eliminate decentralized renewable technologies, suggesting complementarity rather than complete substitution.

**Key Finding 4: Policy Implication**

Future electrification strategies should integrate centralized grids with decentralized renewable technologies instead of treating them as competing systems

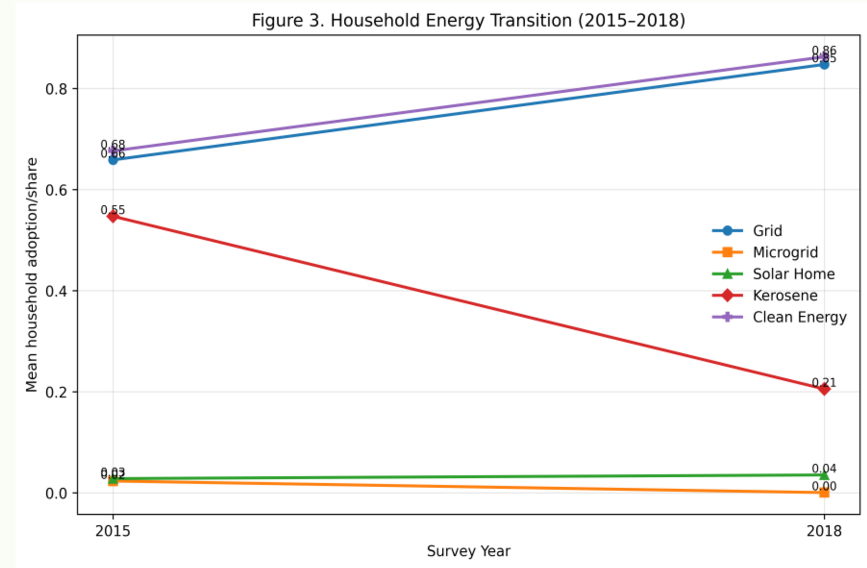



**Figure 4.** Household energy portfolio before and after rapid grid expansion. Grid electricity expanded substantially, kerosene and microgrid use declined, while solar home systems remained relatively stable, indicating selective displacement rather than complete crowding-out.

## 06 CONCLUSIONS

- ✓ Districts experiencing greater grid expansion recorded significant reductions in kerosene use and fossil-based decentralized electricity systems.
- ✓ Renewable decentralized technologies, particularly solar home systems, were not significantly displaced, suggesting selective rather than universal crowding-out.
- ✓ Electrification contributes to a cleaner household energy portfolio, but achieving circular energy transitions ultimately depends on decarbonizing electricity generation.

*"Grid expansion does not crowd out all decentralised energy. It selectively displaces fossil-based systems while allowing renewable technologies to persist."*

## 07 FUTURE WORK

Future research will evaluate long-term complementarities between centralized grids and decentralized renewable systems using post-Saubhagya household surveys.

## 08 REFERENCES

1 Aklin, M., Bayer, P., Harish, S. P., & Urpelainen, J. (2018). Economics of household technology adoption in developing countries: Evidence from solar technology adoption in rural India. Energy Economics, 72, 35–46. https://doi.org/10.1016/j.eneco.2018.02.011

2 Aklin, M., Cheng, C., Urpelainen, J., Ganesan, K., & Jain, A. (2016). Factors affecting household satisfaction with electricity supply in rural India. Nature Energy, 1(11). https://doi.org/10.1038/nenergy.2016.170

3 Bhatia, P. (2023). India's state-led electricity transition: A review of techno-economic, socio-technical and political perspectives. Energy Research & Social Science, 102, 103184. https://doi.org/10.1016/j.erss.2023.103184

## 09 ACKNOWLEDGEMENTS

Mani, S., Shahidi, T., Patnaik, S., Jain, A., Tripathi, S., Ganesan, K., Aklin, M., Urpelainen, J., Chindarkar, N., Council On Energy, Environment And Water, Initiative For Sustainable Energy Policy, & National University Of Singapore. (2019). Access to Clean Cooking Energy and Electricity: Survey of States in India 2018 (ACCESS 2018) (Version 1.0, pp. 10011910, 17582158, 20289049, 266112) [Text/tab-separated-values,text/tab-separated-values,text/tab-separated-values,application/pdf]. Harvard Dataverse. https://doi.org/10.7910/DVN/AHFINM